

Experiment ‘p_2_3_power_functions’ Results

June 3, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_2_3_power_functions

Natural language statement: Write a method that takes a number and returns its square, cube, and fourth power.

Method signature: p_2_3_power_functions(x: int) returns (sq: int, cube: int, fourth: int)

Ensures

- `sq == x * x`
- `cube == x * x * x`
- `fourth == x * x * x * x`

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method that takes a number and returns its square, cube, and fourth power.

The signature should be:

`p_2_3_power_functions(x: int) returns (sq: int, cube: int, fourth: int)`

The method should respect the following contract:

`ensures sq == x * x, ensures cube == x * x * x, ensures fourth == x * x * x * x`

Produce and show only the Dafny body of this method, including the curly braces that
→ surround it. Do not show the signature nor contract. You must not use 'assume' nor
→ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-DmdWqLx2ApIMI5kLQ4WxYxgXpmKDr

Model: gpt-5.5-2026-04-23

Created at: 1780484064

Finish reason: stop

Usage: 318 tokens in, and 116 tokens out

```
{
  sq := x * x;
  cube := x * x * x;
  fourth := x * x * x * x;
}
```

Final Program

```
method p_2_3_power_functions(x: int) returns (sq: int, cube: int, fourth: int)
  ensures sq == x * x
  ensures cube == x * x * x
  ensures fourth == x * x * x * x
{
  sq := x * x;
  cube := x * x * x;
  fourth := x * x * x * x;
}
```

Total Token Usage

Input tokens: 318

Output tokens: 116

Reasoning tokens: 66

Sum of 'total tokens': 434

Experiment Timings

Overall Experiment started at 1780484064334, ended at 1780484070843, lasting 6509ms (6.51 seconds)

Iteration #1 started at 1780484064336, ended at 1780484070843, lasting 6507ms (6.51 seconds)